# ASSIGNMENT (1)

## Part 1: True / False

**1. Expenditure on software represents a significant fraction of the Gross National Product (GNP) in all developed countries.**

**True**

**2. For systems with a long life, development costs are usually several times greater than maintenance costs.**

**False**

**3. Computer science is primarily concerned with the practicalities of developing and delivering useful software, while software engineering focuses on theory and fundamentals.**

**False**

**4. In a customized product, the specification of what the software should do is owned by the software developer.**

**False**

**5. Software reuse is the dominant approach for constructing web-based systems.**

**True**

## Part 2: Multiple Choice (MCQ)

**1. Which of the following is NOT listed as one of the four fundamental software engineering activities?**

**a) Software validation**

**b) Software development**

**c) Software specification**

**d) Software debugging**

**2. Which essential attribute of good software is described as "Software must be acceptable to the type of users for which it is designed"?**

- **a) Maintainability**
- **b) Dependability and security**
- **c) Efficiency**
- **d) Acceptability**

**3. The key challenges facing software engineering are listed as coping with increasing diversity, developing trustworthy software, and...**

- **a) demands for reduced delivery times.**
- **b) competition from overseas.**
- **c) managing hardware costs.**
- **d) finding skilled programmers.**

**4. The personal insulin pump case study is a prime example of which application type?**

- **a) Batch processing system**
- **b) Stand-alone application**
- **c) Embedded control system**
- **d) Entertainment system**

**5. Which ACM/IEEE ethical principle states that "Software engineers shall act consistently with the public interest"?**

- **a) PUBLIC**
- **b) CLIENT AND EMPLOYER**
- **c) PRODUCT**
- **d) JUDGMENT**

## Part 3: Essay Questions

**1. Compare and Contrast Application Types: Using the lecture slides, compare the Embedded Control System (like the insulin pump) with the Interactive Transaction-Based Application (like an e-commerce site). What is the primary purpose of each, and which essential software attribute (Maintainability, Dependability, Efficiency, or Acceptability) do you think is most critical for each? Justify your choices.**

> **Embedded Control System (e.g., Insulin Pump):**
> **Controls hardware in real time for safety-critical tasks.**
> **Most critical attribute: Dependability – because any failure could harm the patient.**
>
> **Interactive Transaction-Based Application (e.g., E-commerce Site):**
> **Handles user transactions like buying or payments.**
> **Most critical attribute: Acceptability – because user satisfaction, usability, and trust are key to success.**

**2. Ethical Dilemma Analysis: The slides describe the MHC-PMS (Mental Health Care Patient Management System), which handles sensitive patient data. One of its key concerns is Privacy. Imagine a manager asks a software engineer to rush the system's release for a deadline, even though testing for data security is incomplete. This could violate the ethical principle of CLIENT AND EMPLOYER (to meet the deadline) but also PUBLIC (protecting patient welfare) and PRODUCT (not meeting professional standards). Briefly discuss this ethical dilemma and explain why it is a difficult choice for the engineer**

**The MHC-PMS manager wants to release the system before security testing to meet a deadline.**
**This conflicts between:**

**Client/Employer: meeting the deadline,**

**Public: protecting patient data,**

**Product: ensuring quality and security.**

**It's difficult because rushing release helps the employer but risks patient privacy.**
**The engineer should prioritize public safety and data protection, even if it delays the release.**